

OFFLINE RESTAURANT POS SOFTWARE

1. Project Overview (This will be a fully offline POS software for restaurants)

- **The software will manage:**
 - Dine-in orders
 - Takeaway orders
 - Tap ordering
 - Kitchen orders (KOT)
 - Billing
 - Inventory / stock
 - Expenses
 - Staff management
 - KDS
 - Scanner for menu
 - E-Invoice (Fatura)
- The software will work **without internet** for normal restaurant operations.
- It will support:
 - Windows
 - macOS
 - Android (for menu)
- There will be **3 pricing plans**:
 - Starter — \$70
 - Growth — \$180
 - Enterprise — \$300
- Each plan will have different features based on the size and needs of the restaurant.

2. Functional Requirements

2.1 Dine-In

- The dine-in module will include:

- Table management
- Table layout
- Table status:
 - Free
 - Occupied
 - Reserved
- Create orders for each table
- Edit orders before billing
- Tap-based ordering
- KOT printing
- Bill generation
- Split bill
- Merge bill
- Table reservations in higher plans

2.2 Tap Ordering

- Orders will be taken using a **touchscreen or tablet**.
- Waiter or cashier can create an order by tapping menu items.
- No manual typing will be required.
- Menu items will be divided into categories, such as:
 - Starters
 - MainCourse
 - Beverages
 - Desserts
- Tapping an item will add it to the order.
- Quantity can be changed using **+ and - buttons**.
- Special instructions can be added if needed.
- Once the order is confirmed, the KOT will be created automatically.

2.3 KOT (Kitchen Order Ticket)

- KOT will be automatically printed on the kitchen printer when an order is confirmed.

KOT will show:

- Table number
- Item name
- Quantity
- Special instructions, if any

Order/token number

- Prices and bill amounts will **not** be shown on the KOT.
- Every KOT will have a unique order/token number for easy tracking.
- If an order is changed after confirmation:
 - New item is added
 - Item is removed
 - Quantity is changed
- The updated KOT will be automatically printed and sent to the kitchen.

2.4 Takeaway

- The takeaway module will include different types of orders.

Walk-in Takeaway

- Customer comes to the restaurant.
- Staff takes the order at the counter.
- Order is added to the system.
- Bill is generated.
- Customer collects the order.

Interchange / Phone Orders

- Customer calls the restaurant.
- Staff takes the order over the phone.

- Order is added to the system.
- Order is prepared for pickup.

House Delivery

- Restaurant uses its own delivery riders.
- Staff creates the order.
- Order is prepared.
- Order is given to the restaurant's own rider.

2.5 HR & Payroll

The HR and payroll module will include:

- Staff attendance
- Staff shifts
- Basic salary calculation
- Leave management
- Salary slips
- Salary deductions
- Bonuses

Advanced HR features will be available in the Enterprise plan.

2.6 KDS (Kitchen Display System)

The KDS module will digitally display kitchen orders on a kitchen screen or tablet.

- Orders will automatically appear on the KDS when confirmed

KDS will show:

- Order / Token number
- Table number
- Order type
- Item name and quantity
- Special instructions
- Order time

Kitchen staff can update order status:

- New
- Preparing
- Ready
- Completed

- Order status will automatically update in the POS system.
- The KDS will work offline through the restaurant's local network

2.7 E-Invoice

The E-Invoice module will generate invoices according to the required tax and billing requirements.

- The system will automatically calculate VAT on invoices.

VAT will support:

- VAT-inclusive pricing
- VAT-exclusive pricing

Each invoice will show:

- Invoice number
- Invoice date and time
- Customer details
- Items and quantities
- Subtotal
- VAT amount
- Grand total
- VAT will be calculated at item level based on the applicable VAT category.
- The system will maintain invoice records for reporting and future reference.
- E-Invoices can be generated and stored digitally

2.8 Customer Loyalty & Points

The system will have a Customer Loyalty Points feature to reward regular customers.

- Staff can create a customer profile with:
 - Customer name
 - Phone number
 - Email (optional)
 - Loyalty points
 - Purchase history
- Customers will earn points when they make a purchase.
- The restaurant can decide how many points a customer earns based on the bill amount.
- For example:
 - \$10 purchase = 10 points
 - \$50 purchase = 50 points
- Points will be automatically added to the customer's account after the bill is completed.

- Customers can use their points to get discounts on future purchases.
- The system will automatically reduce the points when they are used.
- Staff can see the customer's :
 - Total points
 - Available points
 - Used points
 - Purchase history
- The restaurant can set:
 - Points earning rules
 - Points required for discounts
 - Maximum points that can be used on one bill
 - Points expiry, if required
- Loyalty points can be used for:
 - Dine-in orders
 - Takeaway orders
 - House delivery orders
- The system should keep a complete points history so staff can see when points were earned or used.

2.9 Inventory / Stock

Starter Plan

- Basic inventory features:
- Stock in
- Stock out
- Low-stock alerts

Growth Plan

- Additional inventory features:
- Recipe-based stock deduction
- Ingredients are automatically reduced when an item is sold
- Supplier records
- Purchase orders

For example, when 1 burger is sold, the system can automatically reduce:

1 bun

1 patty

1 cheese slice

- Required sauces

Enterprise Plan

- Advanced inventory features:
- Batch tracking
- Expiry date tracking
- Wastage tracking
- Wastage reports
- Wastage analytics
- Automatic reorder suggestions

2.9 Expenses

- Electricity
- Bills
- Staff salary
- Rent
- others

3. Pricing Plans

Starter — \$70

Best for:

Small restaurants

Single outlet

Basic POS requirements

Features

- **Dine-In**
- Full table management
- KOT support
- Tap ordering

- **Takeaway**
- Walk-in takeaway only
- **Inventory**
- Basic stock in/out
- Low-stock alerts
- **HR / Payroll**
- Not included
- **Users / Branches**
- Single store
- 2–3 staff logins

Growth — \$180

- Best for:
- Growing restaurants
- Restaurants with active delivery

Features

Dine-In

Full table management

KOT support

Tap ordering

Takeaway

Walk-in takeaway

- Interchange / phone orders
- House delivery
- **Inventory**
- Recipe-based stock deduction
- Supplier management
- Purchase orders
- **HR / Payroll**
- Staff attendance

- Shift management
- Basic salary calculation
- **Users / Branches**
- Single store
- Multiple users
- Role-based user access

Enterprise — \$300

- Best for:
- Large restaurants
- Restaurant chains
- Multi-branch businesses

Features

Dine-In

Full table management

KOT support

Tap ordering

Table reservations

Takeaway

- Walk-in takeaway
- Interchange / phone orders
- House delivery
- Third-party delivery

- **Inventory**

- Recipe-based stock deduction
- Supplier management
- Purchase orders
- Batch tracking
- Expiry tracking
- Wastage analytics

- **HR / Payroll**

- Attendance
- Shift management
- Leave management
- Salary slips
- Salary deductions
- Bonuses
- **Users / Branches** Multiple branches
 - Custom permissions
 - Role-based access

4. Non-Functional Requirements

Fully Offline

- Core POS operations must work without internet.
- Orders should work offline.
- Billing should work offline.
- Table management should work offline.
- Inventory should work offline.
- Staff management should work offline.

Cross-Platform

- The software should run on:
 - Windows
 - macOS
 - Android (Menu)

Local Data Storage

- Data will be stored locally on the restaurant's computer.
- The system should have backup and restore functionality.
- Data should be easy to backup and restore when needed.

Offline Licensing

- Each device will have a license key.
 - The license can be checked locally.
- Internet should not be required for normal license validation.
- The license can be linked to the specific computer/device.

5. Recommended Tech Stack

- Because this is an offline desktop application and the existing development experience is with MERN, the recommended stack is:

Desktop Application — Electron.js

- Electron.js will be used to create the desktop application.

Frontend — React + Vite

- React + Vite will be used for the user interface.
- It will handle:
 - POS screen
 - Tables
 - Menu
 - Orders
 - Billing
 - Inventory screens
 - HR screens
 - Reports

Backend — Node.js + Express

- The backend will run locally on the customer's computer.
- It will handle:
 - Business logic
 - Orders
 - Billing
 - Inventory
 - HR
 - Printing

- License checking
- Database operations

Local Database — SQLite

- SQLite will be used as the local database.
- Will be use prisma + SQLite
- SQLite is suitable because:
- It works completely offline.
- No separate database server is required.
- Data is stored in a local file.
- easy to backup.
- suitable for a local POS system.

Printing

- Possible options:
- node-thermal-printer
- Native Windows/macOS printing APIs
- Printing will be used for:
- Customer receipts
- KOT
- Kitchen printers

Licensing

- The system can use:
- License key
- Device/hardware fingerprint
- The license can be linked to a specific computer.
- This helps prevent the same license from being used on unlimited devices.

Packaging — electron-builder

- electron-builder will be used to create installers.
- It can create:
- Windows .exe
- macOS .dmg

- The same project can be used to build both versions.

6. Licensing & Distribution

- Each pricing plan can be controlled using **feature flags**.
- For example:
- Starter license → Starter features are unlocked.
- Growth license → Growth features are unlocked.
- Enterprise license → All features are unlocked.
- The main software/codebase will remain the same.
- Only the features available to the customer will change based on their license.

7. Backup & Restore

- Because the system is offline, local data backup is very important.
- The software should provide:
- Manual backup
- Restore backup
- Backup database to another location
- Option to save backup on USB or external storage
- Automatic backup option if required
- This will help protect restaurant data if the computer has a problem.

Budget

Developing cost would be: 500\$